

Asymptotics of Partition Functions on Coulomb Gases

João Victor Alcantara Pimenta,
Guilherme L. F. Silva
Universidade de São Paulo
joaovictorpimenta@usp.br



XV Workshop de Teses e Dissertações em Matemática

The theory of random matrices finds its way in a diverse range of areas of Mathematics and Physics, and in particular in statistical mechanics. In the study of the spectra of eigenvalues of some random matrix ensembles, a physical analogy regarding thermodynamical systems becomes evident and the use of physical motivations in its study is of great importance. To better understand these systems, we study a known concept for physicists - the so-called partition functions for Coulomb Gases. These objects are of fundamental importance in the pursuit of understanding models of random matrices - thought of as particle systems composed of eigenvalues. In particular, we are concerned about random matrices of large size. Our goal is then to have a good understanding of the asymptotic of the partition function for the given large system. Recently, asymptotics for some classes of potentials for two dimensional Coulomb Gases have been calculated using a novel method based on asymptotics of orthogonal polynomials, a much more straightforward approach that enables us to explore the partition functions for other systems of interest. In light of this, we will properly understand this result and look for possible models where we can reapply the technique for new results. We start our study with an important class of examples which have its asymptotics expressed in terms of Hermite functions. To study them, we use their integral representation, going through a better understanding of classical methods for asymptotics of integrals, such as the Watson Lemma, Laplace Method and Steepest Descent, to name a few. Besides characterizing these asymptotics, we understand its place in the theory of random matrix and search for new results the technique could provide.

Density Function of Matrix Models

Invariant ensembles of random matrices, that we are interested in, are the ones in which the probability distribution is expressed as

$$p(\hat{M})d\hat{M} = \frac{1}{Z_N} e^{-\text{Tr}(V(\hat{M}))} |\Delta(\Lambda)|^\beta d\Lambda dU,$$

for any polynomial potential $V : \mathbb{R} \rightarrow \mathbb{R}$ such that $V(x) \rightarrow \infty$ sufficiently rapidly as $|x| \rightarrow \infty$. Here, we also considered Weyl's formula to express $d\hat{M} = |\Delta(\Lambda)|^\beta d\Lambda dU$. We note that $|\Delta(\Lambda)|$ is the jacobian expressed as a Vandermonde determinant, $d\Lambda$ is the Lebesgue measure on $\mathbb{R}^N$, dU is the Haar measure on the set of invertible matrices and β is the inverse temperature, which we take to be 2 for the rest of this work. As this ensemble is invariant, we can express the jpdf of the random matrix as a function on the eigenvalues

$$p_{ord}(\lambda_1, \lambda_2, \dots, \lambda_N) = \frac{1}{Z_{N,\beta}^{(ord)}} \prod_{i=1}^N e^{-V(\lambda_i)} \prod_{i < j} |\lambda_i - \lambda_j|^2. \quad (1)$$

This has an interesting interpretation as a Coulomb gas. The measure models an interacting gas of indistinguishable particles under the influence of an external potential at positions $\lambda_1, \lambda_2, \dots, \lambda_N \in \mathbb{C}$ in an external ambient of two dimensions.

The Kernel Representation

There is a further manipulation we could do on the expression given by 1. Namely, for $\beta = 2$

$$\begin{aligned} p_{ord}(\lambda_1, \lambda_2, \dots, \lambda_N) &= \frac{1}{Z_{N,\beta}^{(ord)}} \left[\det(m_i^{j-1})_{i,j=1,\dots,N}^2 \right] \prod_{i=1}^N e^{V(\lambda_i)} \\ &= \frac{1}{Z_{N,\beta}^{(ord)}} \left[\det(p_{j-1}(\lambda_i))_{N \times N} \right]^2 \prod_{i=1}^N e^{V(\lambda_i)} \\ &= \frac{1}{Z_{N,\beta}^{(ord)}} \det(K_N(\lambda_i, \lambda_j)) \prod_{i=1}^N e^{V(\lambda_i)}, \end{aligned}$$

where $K_N(x, y) = \sum_{j=0}^{N-1} p_j(x)p_j(y)$. If we choose p_j to be the Hermite polynomial, by algebraic properties

$$\begin{aligned} \hat{K}_N(x, y) &= e^{-\frac{V(x)}{2}} e^{-\frac{V(y)}{2}} K_N(x, y) \\ &= e^{-\frac{V(x)+V(y)}{2}} \frac{H_N(x)H_{N-1}(y) - H_{N-1}(x)H_N(y)}{x - y}. \end{aligned} \quad (2)$$

This motivate the study of asymptotics of orthogonal polynomials, especially the Hermite polynomial.

Asymptotics Methods

We now focus on Hermite polynomials. It admits the integral representation

$$H_n(x) = \frac{n!}{2\pi i} \int_{\Gamma} z^{-n-1} e^{2xz-z^2} dz =: c_{\lambda} \int_{\gamma} g(s) e^{-\lambda f(s)} ds = \phi(\lambda) \quad (3)$$

where Γ is a circle around the origin for the Hermite polynomial.

Theorem 1 (Steepest Descent) Take γ a smooth contour. Assume that the exponent $f = u + iv$ is an analytic function on a neighborhood of Γ that satisfies $v(z) \equiv v_0, z \in \gamma$. In addition, suppose that u has a unique minimum at an interior point of Γ , say on $z_0 \in \gamma$, with $f''(z_0) \neq 0$, and also that g is analytic on a neighborhood of γ and satisfies $\int_{\gamma} |g(z)||dz| < \infty$. Then ϕ has a leading asymptotic given by

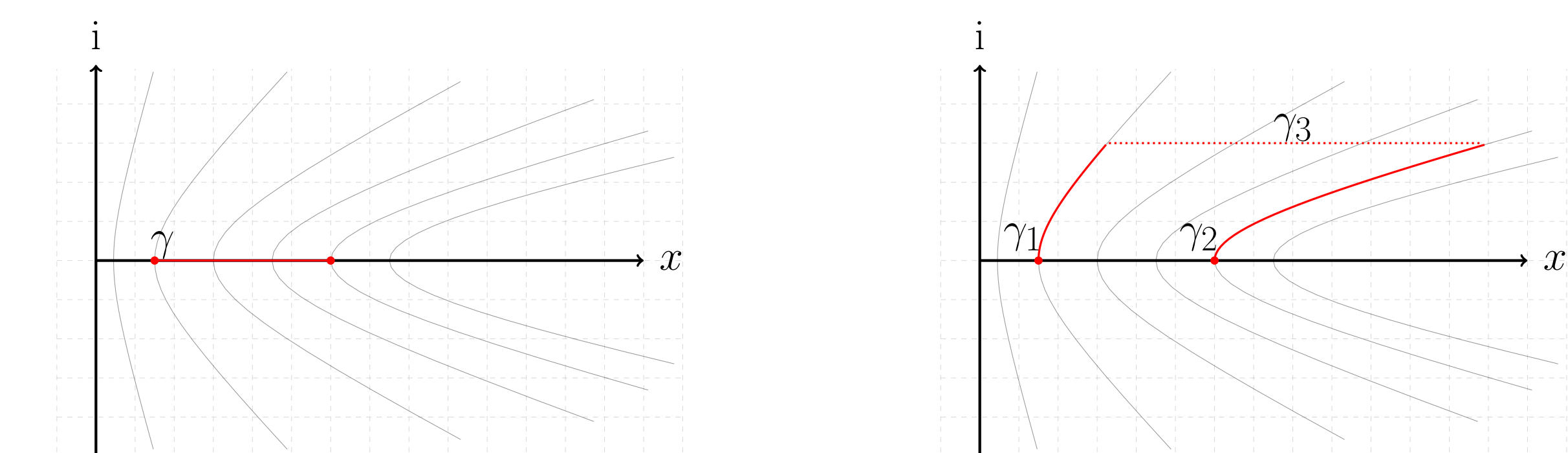
$$\phi(\lambda) = e^{-\lambda f(z_0)} \left(\frac{\sqrt{2\pi} g(z_0) e^{i\theta(z_0)}}{\lambda^{1/2} \sqrt{|f''(z_0)|}} + \mathcal{O}(\lambda^{-3/2}) \right), \quad \lambda \rightarrow +\infty, \quad (4)$$

where $\theta(z_0)$ is the angle of the oriented tangent of γ at z_0 .

The assumption of analytic functions, allows the deformation of the contour. To choose an appropriate contour, note that the gradient for the imaginary and real part are orthogonal at any point, i.e.

$$\langle \nabla u(z), \nabla v(z) \rangle = u_x(z)v_x(z) + u_y(z)v_y(z) = u_x(z)v_x(z) - v_x(z)u_x(z) = 0.$$

Taking the critical points of f we may find a path passing through where the imaginary part is constant, this will follow the *Steepest Descent* path of the real part.



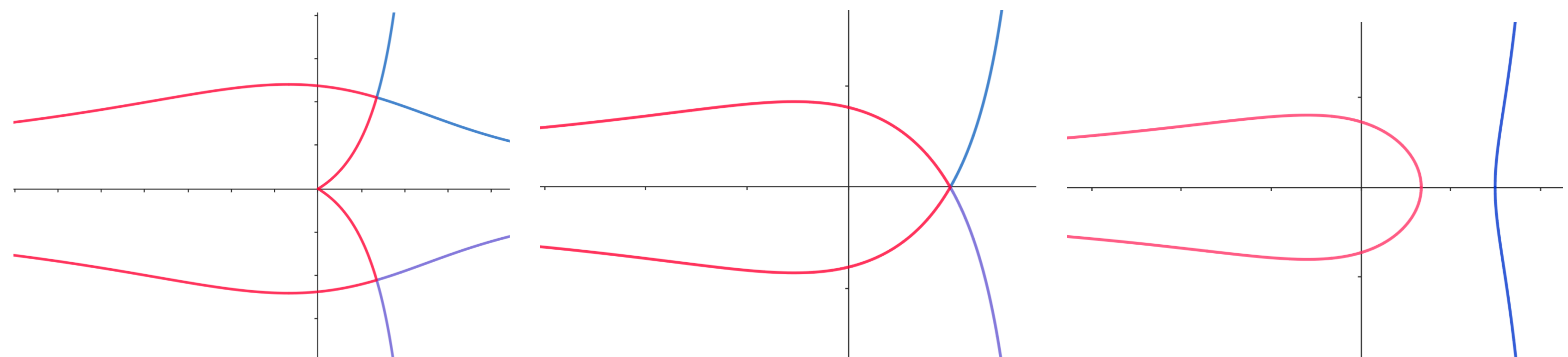
The problem reduces to the evaluation of real integrals. We note that, in the limit, the integrand accumulates around the critical points. Theorem 1 gives an estimative for the integral, given the critical point.

Hermite Polynomials

To use the result on the Hermite polynomials, as in 3, we first consider the scaling $x \mapsto \sqrt{n}\alpha$ and $z \mapsto \sqrt{n}\beta$

$$H_n(\sqrt{n}\alpha) = C_n^0 \int_{\Gamma} e^{-n(\phi_{\alpha}(\beta))} \beta^{-1} d\beta, \quad H_{n-1}(\sqrt{n}\alpha) = C_n^1 \int_{\Gamma} e^{-n(\phi_{\alpha}(\beta))} d\beta,$$

where $\phi_{\alpha}(\beta) = -2\alpha\beta + \beta^2 + \log(\beta)$. We determine the critical points $\beta_c^{\pm} = \frac{\alpha \pm \sqrt{\alpha^2 - 2}}{2}$ as solutions to $\phi'_{\alpha}(\beta) = 0$. Because of the critical point behaviour, we divide the problem into three cases: $0 < \alpha < \sqrt{2}$, $\alpha = \sqrt{2}$ and $\alpha > \sqrt{2}$ and solve for $\text{Im}(\phi_{\alpha}(z) - \phi_{\alpha}(\beta_c^{\pm})) = 0$. The solution to this equation are the paths shown below



We now apply Laplace Method for each deformation on the determined paths of *Steepest Descent*, determining the asymptotic related to each case. We show results for H_n but the same is done for H_{n-1} .

- For $0 < \alpha < \sqrt{2}$: Γ deformed to $\Gamma_d^+ \cup \Gamma_d^- \cup \Gamma_x$, $\alpha = \sqrt{2} \cos(\gamma)$ and $\beta_c^{\pm} = \sqrt{2}/2 e^{\pm i\gamma}$

$$H_n(\sqrt{2n} \cos(\gamma)) \approx \frac{n! n^{-\frac{1}{2}(n+1)}}{2 \sin(\gamma) \sqrt{\pi}} e^{-n(-\cos^2(\gamma) - \frac{1}{2} + \ln \frac{\sqrt{2}}{2})} \cos(n\theta_{\gamma} - \theta(\beta_c^+))$$

- For $\alpha > \sqrt{2}$: Γ deformed to $\Gamma_{\beta_c^-}$, $\alpha = \sqrt{2} \cosh(\gamma)$ and $\beta_c^{\pm} = \sqrt{2}/2 e^{\pm \gamma}$

$$H_n(\sqrt{2n} \cosh(\gamma)) \approx \frac{n! n^{-\frac{1}{2}(n+1)} e^{\gamma}}{\sqrt{2\pi} |1 - e^{2\gamma}|} e^{-n(-2 \cosh(\gamma) e^{-\gamma} + \frac{1}{2} e^{-2\gamma} + \gamma + \ln(\frac{\sqrt{2}}{2}))}.$$

Conclusion

Given the well-established importance of random matrices, we explore its properties. Using techniques of deformation of path of integration on complex integrals and estimatives of real integrals with critical points under some limit, we determine the asymptotics of Hermite Polynomials H_n . With this result, we can also determine asymptotics on the so-called kernel K_n . This object has great importance as it shows up in the expressions of the equilibrium measures for models of random matrices. Therefore, we can study properties of such models related to these measures. This is the next step in the project: asymptotic analysis of partition functions on Coulomb gases, an analogue of models of random matrix, for settings of interest.

Acknowledgements

We would like to thank the São Paulo Research Foundation (FAPESP) for Financial Support under Process Number 2024/06997-1. We also thank the Institute of Mathematical and Computer Sciences (ICMC) for the Institutional Support under the PPG-MAT masters program.

References

- [1] Mehta, M., Gaudin M., *On the density of eigenvalues of a random matrix*, Nuclear Physics, **1960**.
- [2] Dyson, F., *Statistical theory of the energy levels of complex systems*, Journal of Mathematical Physics, **1962**.
- [3] Nico M. Temme, *Asymptotic Methods for Integrals*, World Scientific Publishing Company, **2014**.
- [4] Potters, M., Bouchaud, J., *A First Course in Random Matrix Theory: for Physicists, Engineers and Data Scientists*, Cambridge University Press, **2020**.
- [5] Sung-Soo Byun, Nam-Gyu Kang, Seong-Mi Seo *Partition Functions of Determinantal and Pfaffian Coulomb Gases with Radially Symmetric Potentials*, Springer Science and Business Media LLC, **2023**.